

VPA IB DP CS — IGCSE to IB DP Bridge Guide

IGCSE to IB DP Bridge Guide 从IGCSE到IB DP过渡指南 Cóng IGCSE dào IB DP Guòdù Zhǐnán

From: IGCSE Computer Science (0478) | To: IB DP Computer Science (First Assessment 2024)

Introduction

VPA students enter IB DP Computer Science with strong foundations from IGCSE 0478. This guide maps the transition, identifies knowledge gaps, and provides targeted preparation to ensure every student starts Year 1 confident and ready.

1. Topic Mapping: IGCSE → IB DP

IGCSE 0478 Topic	IB DP Topic	Transition Notes
1.1 Data representation (binary, hex, ASCII, images, sound)	2.4 Binary representation + 2.2 Memory	IB adds: floating-point, two's complement, truncation errors, compression
1.2 Data transmission (packets, errors, protocols)	3.1 Network fundamentals	IB adds: TCP/IP layers, DNS, DHCP, packet switching depth
1.3 Hardware (CPU, memory, storage, I/O)	2.1 Computer architecture + 2.2 Memory & storage	IB adds: FDE cycle detail, registers, cache, virtual memory, OS functions
1.4 Software (OS, utilities, applications)	2.3 Operating systems & application software	IB adds: software licensing, types of OS in depth
1.5 Security (malware, hacking, prevention)	3.3 Network security	IB adds: encryption algorithms, SSL/TLS, PKI, firewalls in depth
2.1 Algorithm design (pseudocode, flowcharts)	4.1 Programming fundamentals	IB adds: IB-specific pseudocode conventions, collections, file handling
2.2 Programming (Python, variables, loops, IF)	4.1 Programming fundamentals	IB adds: procedures/functions, parameters, scope, array handling
2.3 Databases (tables, queries, forms, reports)	1.2 System design (ER diagrams)	IB adds: ER modelling, cardinality, normalisation concepts
2.4 Boolean logic (AND, OR, NOT)	2.5 Simple logic gates	IB adds: NAND, NOR, XOR, circuit construction, truth tables for all gates
N/A	1.1 Systems in organizations	NEW: Feasibility, changeover, project management, stakeholders
N/A	4.2 Data structures	NEW: Records, arrays of records, collections, stacks, queues
N/A	4.3 Algorithms	NEW: Bubble, selection, insertion, merge sort; linear/binary search; recursion

IGCSE 0478 Topic	IB DP Topic	Transition Notes
N/A	4.4 Software development	NEW: Testing types, debugging, documentation, trace tables
N/A	5.1 – 5.3 Abstract Data Structures	HL ONLY: Linked lists, trees, graphs, dynamic allocation
N/A	6.1 – 6.2 Resource Management	HL ONLY: Scheduling, paging, deadlock, control systems
N/A	7.1 – 7.3 Control	HL ONLY: Centralized, distributed, autonomous systems

2. Knowledge Gaps to Address

Gap 1: Systems Thinking (Topic 1)

IGCSE focuses on technical components. IB DP Topic 1 requires organisational understanding: stakeholders, feasibility, changeover, project management. Students must shift from "how does this work?" to "how do we implement this in an organisation?"

Preparation Tasks:

Interview a staff member about a system change at VPA

Read a case study of a failed IT implementation (e.g., NHS IT project)

Define: stakeholder, feasibility, milestone, changeover in your own words

Gap 2: Binary Depth (Topic 2.4)

IGCSE covers binary, hex, and basic ASCII. IB DP adds:

Two's complement for negative numbers

Floating-point representation (mantissa and exponent)

Truncation and rounding errors

Binary shifts (logical and arithmetic)

Gap 3: Logic Gates (Topic 2.5)

IGCSE covers AND, OR, NOT. IB DP requires:

NAND, NOR, XOR, XNOR

Constructing circuits from Boolean expressions

Deriving expressions from truth tables

Simple applications (half-adders, security systems)

Gap 4: Algorithm Rigor (Topic 4)

IGCSE programming is practical. IB DP requires:

Standard IB pseudocode (not Python-specific)

Trace tables for all algorithms

Efficiency analysis (Big O at HL)

3. Summer Preparation Programme (Pre-Year 1)

Week	Focus	Tasks	Time
1	Binary Mastery	Two's complement practice; binary shifts; floating-point introduction	3 hrs
2	Logic Gates	Learn NAND, NOR, XOR; complete truth tables; build simple circuits	3 hrs
3	IB Pseudocode	Translate Python programs to IB pseudocode; practice conventions	3 hrs
4	Systems Thinking	Read about a real system implementation; identify stakeholders and risks	2 hrs
5	Sorting & Searching	Learn bubble sort and linear search; trace through examples	3 hrs
6	Review & Consolidation	Mixed practice test; identify remaining weak areas	2 hrs

4. Differentiated Entry Points

Level	IGCSE Grade	Starting Focus	Target
Bronze	C - D	Secure IGCSE fundamentals; master IB pseudocode; build confidence	IB grade 4 - 5
Silver	B - A	Extend to IB depth; practise exam technique; begin IA thinking	IB grade 5 - 6
Gold	A* - A	HL preparation; advanced reading; mentor Bronze students; IA client identified	IB grade 6 - 7

5. Resources for Bridging

Computer Science Illuminated (Dale & Lewis) — Chapters 1-4, 6-8

Cambridge IB Computer Science Coursebook (Watson & Reyner) — Topic 1 and 2

VPA Bilingual Glossary — Start reviewing 10 terms per week

IB Pseudocode Reference — Available on VPA CS portal

Codecademy Python — Review for programming confidence

CS50's Introduction to Computer Science (Harvard) — Free on edX

6. Key Mindset Shifts

IGCSE Mindset	IB DP Mindset
"What is the right answer?"	"What are the assumptions behind this answer?"
"Does the code work?"	"Is the code efficient, testable, and well-documented?"
"Learn the facts"	"Evaluate the arguments and justify conclusions"
"Programming is about syntax"	"Programming is about solving problems for people"
"Technology is neutral"	"Technology has ethical and social implications"

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